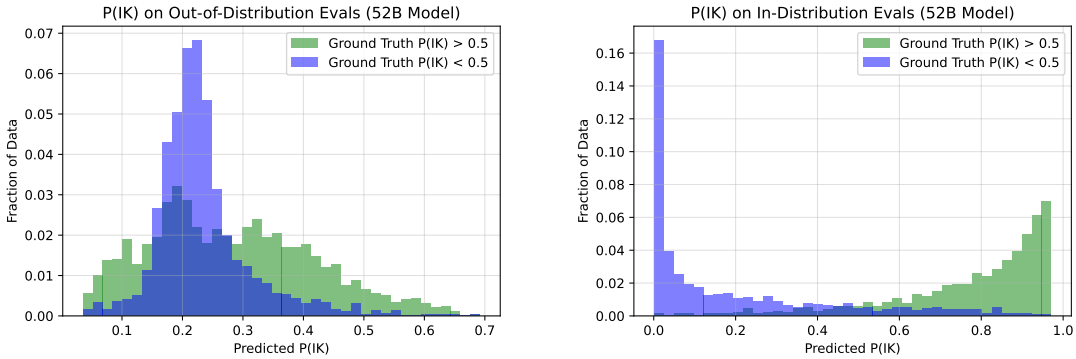

Figure 2 **Left**: We train a P(IK) classifier to predict whether or not a model knows the answer to TriviaQA questions, and then evaluate on Arithmetic, Python Function Synthesis, and Lambada questions. This histogram shows P(IK) scores exclusively from OOD questions. **Right**: We train a P(IK) classifier on TriviaQA, Arithmetic, Python Function Synthesis, and Lambada and histogram P(IK) scores for their test sets.

We find that language models can easily learn to perform well at evaluating P(IK), the probability that they know the answer to a question, on a given distribution (see Figure 3 for an illustration). More intriguingly, we also find some generalization across tasks, for example from trivia to story completion, math, and code, though models struggle with calibration OOD. We also observe some other types of generalization: although P(IK) was only trained on bare questions, it generalizes in such a way that it increases when we provide source materials that address these questions (for trivia) and when we provide hints for math word problems.

1.1 Contributions

Calibration: Multiple Choice, None of the Above, and True/False

- We show that when we use a format with visible lettered answer options, large models are very well calibrated on diverse multiple choice questions (e.g. from BIG Bench [Srivastava et al., 2022], MMLU [Hendrycks et al., 2021], and many other evaluations); see Figures 4, 5, and 6.
- Calibration improves with model size, and it also improves when we pass from zero-shot to few-shot evaluation; see Figure 4.
- Replacing an option with 'none of the above' reduces accuracy and calibration significantly with our models (see Figure 7). However, our models are also well calibrated on True/False distinctions (see Figure 8), with accuracy and calibration also increasing with model capability.
- We also show that RLHF policies [Bai et al., 2022] naively seem miscalibrated, but with a simple temperature adjustment they become fairly well-calibrated on several evaluations (Figure 9).

Self-Evaluation of Model-Generated *Samples*, without Finetuning

We evaluate on the generative tasks TriviaQA [Joshi et al., 2017], Lambada [Paperno et al., 2016], GSM8k [Cobbe et al., 2021], the Codex HumanEval [Chen et al., 2021], arithmetic problems, and natural function synthesis problems scraped from GitHub. See Figure 1 for a brief overview of results.

- Models can self-evaluate whether their own samples are True or False, though this tends to be a more challenging task (since models tend to find their own samples more plausible). Self-evaluations are well-calibrated few-shot, though models aren't as well-calibrated zero-shot. In particular, larger k for k-shot self-evaluation seems to primarily help by improving calibration, rather than by improving the AUROC for separating correct and incorrect responses. See Figures 10 and 11 in the main text for representative results, and Figures 28, 29, and 30 in the appendix for complete results.
- Showing models many of their own $T = 1$ samples, along with a single sample to evaluate as True/-False, can significantly improve their performance (this is somewhat reminiscent of self-consistency prompting [Wang et al., 2022]).

We conclude that language models can perform well at this task few-shot, with most measures of performance improving with model size, even though models are being asked to evaluate their own samples.

Finetuning to Identify the *Questions* Models Can Correctly Answer

We are also interested in whether models know, or can be taught, to specifically identify *questions* that they can and cannot answer, rather than simply evaluating whether answers to questions are in fact correct. Figure 2 provides a brief overview of results.

- We train models with a value head to predict the probability that they can answer a question correctly, which we refer to as P(IK). We find that models trained on TriviaQA have significant power to differentiate between math, lambada, and code questions that they can answer; see Figure 12. However, P(IK) tends to be poorly calibrated on these other distributions (see Figure 13). Generalization is perhaps most clearly illustrated in Figures 16 and 17.
- We study generalization of P(IK) to the inclusion of source materials and mathematical hints, see Figures 18 and 19. We find that P(IK) responds appropriately to sources, correct hints, and incorrect or distracting hints.
- We compare two models with roughly equal capability, but different pretraining data, in Figure 20. These models are somewhat better at predicting their own P(IK) rather than that for another model. We also try "crossing" the P(IK) training data for these models, and find a small effect suggesting that models generalize better when trained on what they know, rather than on what the other model knew.

Thus we conclude that we only find partial generalization on this task.

Glossary: Observables and Metrics

- **P(True)** – The probability a model assigns to the proposition that a specific sample is the correct answer to a question.
- **P(IK)** – The probability a model assigns to "I know", i.e. the proposition that it will answer a given question correctly when samples are generated at unit temperature. In this work, P(IK) is usually computed using a binary classification head on top of a language model.
- **Ground Truth P(IK)** – The fraction of unit temperature samples to a question that are correct.
- **Calibration Charts** - We often plot prediction probability vs frequency that a prediction was correct, see Figure 4 as an example. We use all predictions (not just predictions for the correct answer) and put the same number of predictions in each bin (rather than using equally spaced bins).
- **Expected or RMS Calibration Error** - To obtain a single number summarizing calibration, we compute the ECE as the mean of the absolute difference between probabilistic predictions and frequencies. For ECE we always use 10 bins, with an equal number of predictions in each. What we call the ECE has also been called the mean absolute deviation calibration error [Lin et al., 2022]. To more closely match other works, we only use the most likely predictions (for multiple choice) when computing the ECE, instead of using all predictions. We also include the RMS calibration error, which is better motivated theoretically, but seems to be less widely used in the literature.
- **AUROC** - We sometimes share the area under the receiver operating characteristic (AUROC) discriminating between questions models do or do not know the answer to, or discriminating between samples a model does or does not identify as correct. This captures discriminative power but is indifferent to calibration (note chance AUROC is 0.5, and larger scores are better).
- **(Conditional) Accuracy** - When models self-evaluate the probability P(True) that their own samples are valid, we are very interested in the accuracy of the samples that models label as 'True' (i.e. correct), and how this compares to the accuracy on the full distribution of problems in the task.
- **Brier Score** - In some cases we observe tradeoffs between discrimination (e.g. best AUROC) and calibration. Brier scores combine the discriminative power of self-evaluation with calibration (note the chance Brier score on binary choice tasks is 0.25, and smaller scores are better).

1.2 Models and Evaluation Tasks

Our goal in this study is to evaluate calibration and generalization on a diverse range of tasks. As such we include all of the multiple choice evaluations in BIG Bench [Srivastava et al., 2022], the MMLU

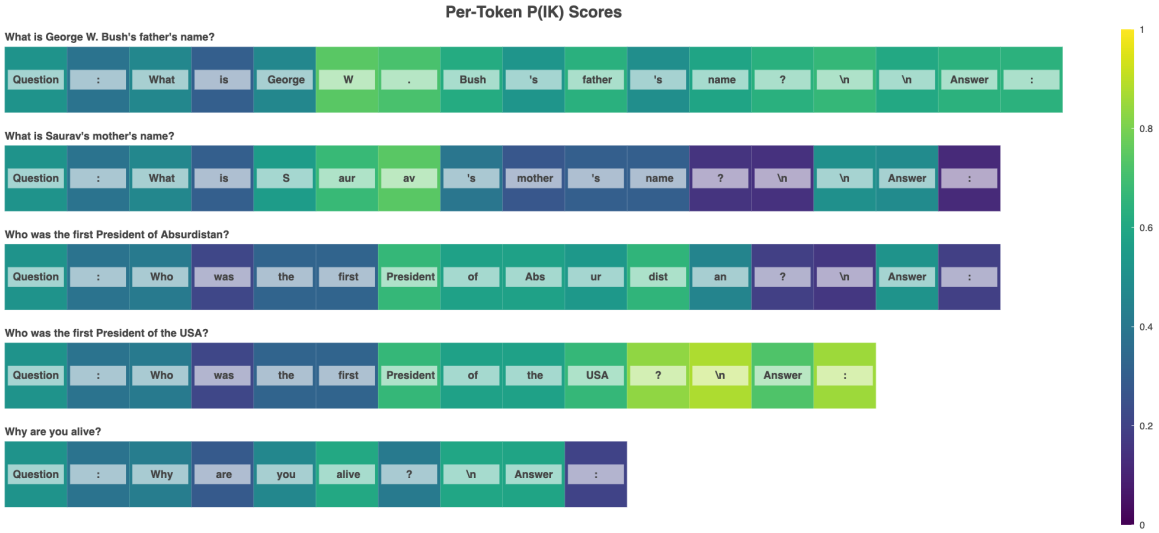

Figure 3 Examples of P(IK) scores from a 52B model. Token sequences that ask harder questions have lower P(IK) scores on the last token. To evaluate P(IK) on a specific full sequence, we simply take the P(IK) score at the last token. Note that we only train P(IK) on final tokens (and not on partial questions).

evaluation [Hendrycks et al., 2021], TruthfulQA [Lin et al., 2021], LogiQA [Liu et al., 2020], and QuALITY [Pang et al., 2021]. We are most interested in open-ended generation, so we study the sampling-based evaluations TriviaQA [Joshi et al., 2017], Lambada [Paperno et al., 2016], the Codex HumanEval [Chen et al., 2021], GSM8k [Cobbe et al., 2021], some basic arithmetic problems, and a dataset of additional web-scraped Python function synthesis problems. See Appendix D for more information on our arithmetic and function synthesis datasets. Ultimately, we would like to train models to express calibrated confidence levels when generating long-form responses in natural language dialogue.

When we perform few-shot evaluation, we simply stuff the context with randomly chosen examples from the (test) evaluation itself. For BIG Bench this means all few-shot examples come from within the specific subtask we are evaluating, though in the case of MMLU we randomize across the entire evaluation, without respecting subject boundaries.

We study a series of language models with 800M, 3B, 12B, 52B parameters. We do not include smaller models because they perform poorly on many of the evaluations we consider. The architecture and training setup for these models is identical to that in [Bai et al., 2022], except that the models we consider here were pretrained for 850B tokens, rather than the 400B tokens used in that work. For simplicity, we do not study models that have been finetuned on python code, though our pretraining distribution includes about 10% code. In section 3.3 we briefly study helpful and harmless RLHF policies finetuned (via the process in [Bai et al., 2022]) from these language models, but otherwise we only study pure language models. We show accuracies for our models on a few of the multiple choice tasks we study in Figure 37 in the appendix.

1.3 Related Work

Calibration for general ML predictions, and interventions to improve calibration, have been studied [Nguyen and O'Connor, 2015, Hendrycks and Gimpel, 2016, Nalisnick et al., 2019, Guo et al., 2017, Hendrycks et al., 2018, Ovadia et al., 2019, Minderer et al., 2021] for some time. Calibration for language models and QA has also been studied [Desai and Durrett, 2020, Jiang et al., 2021], but typically it has been found that to achieve good calibration predictions must be adjusted. Selective prediction, where models abstain from answering certain questions, has been studied as well [Varshney et al., 2022]. Recently, the calibration of a wide range of models was analyzed on the diverse BIG Bench suite of tasks [Srivastava et al., 2022], where it was shown that language model calibration improves with model size. We are indebted to the BIG Bench collaboration for providing a convenient, huge, and diverse evaluation set. The authors of Gopher [Rae et al., 2021] briefly studied calibration on MMLU [Hendrycks et al., 2021] and found promising results, which led us to experiment with a variety of multiple choice formats.